

Lesson 4.2 Analysis of Repeated Measures Designs

Introduction

In statistics and research, experiments with repeated measurements (often called **Repeated Measures Designs**) are studies where the same subjects are measured multiple times under different conditions or over a period of time.

For example, in a standard “between-subjects” design, you might give 10 people Coffee and 10 different people Water to see who is more alert. In a repeated measures design, you give all 20 people Water on Monday and then give those same 20 people Coffee on Tuesday.

In clinical psychology, the experimenter measures the anxiety levels of patients before, during, and after therapy.

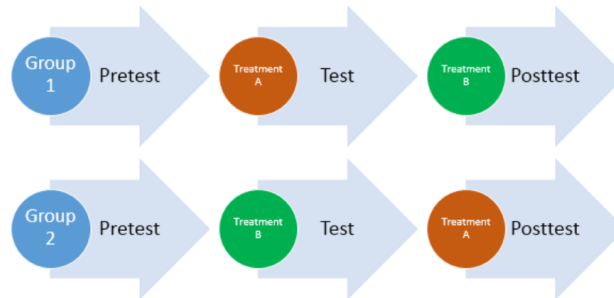
Key characteristics

1. *The Subject is their own Control*: Because you’re measuring the same person twice, you naturally account for their individual quirks (like their baseline caffeine tolerance).
2. *Sequential Timing*: Measurements are taken at multiple time points (e.g., Pre-test, Post-test, and Follow-up). Instead of comparing Person A to Person B, we compare Person A at Time 1 to Person A at Time 2.
3. *Reduced Noise*: This design is statistically powerful because it eliminates “inter-subject variability”—the random differences between different people.

But note that the measurements are correlated and violate the statistical assumption of independence.

Common Types

1. **Pre-test/Post-test**: Measuring a baseline, applying a treatment, and measuring again. This is the simplest form of a repeated measures design. As an example, consider testing a student’s math skills before and after a 4-week tutoring program.



2. **Longitudinal:** Measuring the same variable in the same people over months or years. Tracking the reading level of a group of children from 1st through 5th grade. In agriculture, we measure plant height for 8 weeks.
3. **Crossover:** Participants receive different treatments in a random order. A patient tries Drug A for a week, has a “washout” period, then tries Drug B.

The Benefits

1. Efficiency: You need fewer participants to get statistically significant results.
2. Sensitivity: It is much easier to detect small effects because you aren’t distracted by the differences between individuals.

The Challenges

1. Order Effects: If a participant does Task A then Task B, they might do better on B simply because they practiced during A.
2. Boredom/Fatigue: Participants might get tired or lose interest by the third or fourth measurement.
3. Attrition: If your study lasts a year, some people will inevitably move away or stop responding to your emails.
4. Carryover Effects: The effect of “Treatment A” might still be in their system when they start “Treatment B.”

The Solution: Counterbalancing

To fix the risks above, researchers use Counterbalancing—varying the order of conditions so that “order effects” cancel each other out.

1. Full Counterbalancing: All possible orders are used.
2. Latin Square Design: A simplified version where each condition appears in each “slot” (1st, 2nd, 3rd) at least once.

Assumptions

1. Normality

The dependent variable should be approximately normally distributed at each level of the within-subjects factor. This can be tested using the Shapiro-Wilk test or a Q-Q plot for each time point can be generated. Luckily, analysis of variance is “robust” to minor violations of normality, meaning it still works well if the data is only slightly skewed.

2. Sphericity

This is the most critical assumption for repeated measures. This means that the variances of the differences between all possible pairs of within-subject conditions must be equal. This means that the variances between any pair of repeated measurements

$$Var(Time1 - Time2) \approx Var(Time2 - Time3) \approx Var(Time1 - Time3)$$

This can be tested using Mauchly's test. If $p < .05$, use the Greenhouse-Geisser correction.

3. Absence of Outliers

Repeated measures ANOVA is highly sensitive to outliers. A single extreme measurement for one plant at one time point can significantly skew the mean and increase the error term. This can be checked using box plots.

An Example

The effect of three treatments on the growth of a plant were evaluated in an experiment. One of the parameters is plant height (HT) which is measured 15, 30, 45, and 60 days after transplanting. The experiment was laid out in RCBD.

```
library(tidyverse)
library(rstatix)
library(readxl)
df_wide <- read_excel("agrondata.xlsx")

df_wide <- df_wide %>%
  select(treatment, block, HT15, HT30, HT45, HT60) %>%
  mutate(Block=factor(block),
         Trt=factor(treatment))
head(df_wide)
```

```
## # A tibble: 6 x 8
##   treatment block HT15 HT30 HT45 HT60 Block Trt
##   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <fct> <fct>
## 1      1      1  56.2 102.  152.  210. 1      1
## 2      1      2  53.9  92.9  140   194. 2      1
## 3      1      3  53.4  90.4  134.  189. 3      1
## 4      1      4  52.2  89    129.  183. 4      1
## 5      1      5  54.6 102.  150.  207. 5      1
## 6      2      1  47.3  73.5  131.  173. 1      2
```

```
#Convert to long format
df_long <- df_wide %>%
  pivot_longer(cols = starts_with("HT"),
               names_to = "time",
               values_to = "height") %>%
  mutate(time = factor(time, levels = c("HT15", "HT30", "HT45", "HT60"))) %>%
  select(-c("treatment", "block")) %>%
  mutate(unit_id = interaction(Trt, Block))

head(df_long)
```

```
## # A tibble: 6 x 5
##   Block Trt   time height unit_id
##   <fct> <fct> <fct>   <dbl> <fct>
## 1 1      1    HT15    56.2 1.1
## 2 1      1    HT30   102. 1.1
## 3 1      1    HT45   152. 1.1
## 4 1      1    HT60   210. 1.1
## 5 2      1    HT15    53.9 1.2
## 6 2      1    HT30    92.9 1.2
```

```
res.aov <- anova_test(           # From the rstatix package
  data = df_long,
  dv = height,
  wid = unit_id,
  within = time,                 # The repeated factor
  between = Trt                  # The main factor
)
```

Results of the Mauchly's test is shown below. The results indicate that the sphericity assumption is violated.

```
# Extracting the Mauchly's test results and the Greenhouse-Geisser correction
res.aov$`Mauchly's Test for Sphericity`
```

```
##      Effect      W      p p<.05
## 1      time 0.147 0.001      *
## 2 Trt:time 0.147 0.001      *
```

```
get_anova_table(res.aov)
```

```
## ANOVA Table (type II tests)
##
##      Effect  DFn  DFd      F      p p<.05  ges
## 1      Trt  2.00 12.00  244.010 1.91e-10      * 0.969
## 2      time  1.44 17.28 2301.339 7.63e-21      * 0.979
## 3 Trt:time  2.88 17.28  180.715 3.73e-13      * 0.877
```

Since the sphericity assumption is violated we will report the p-value of the Greenhouse-Geisser correction

```
res.aov$`Sphericity Corrections`
```

```
##      Effect  GGe      DF[GG]      p[GG] p[GG]<.05  HFe      DF[HF]      p[HF]
## 1      time  0.48 1.44, 17.28 7.63e-21      * 0.528 1.58, 19.01 1.02e-22
## 2 Trt:time  0.48 2.88, 17.28 3.73e-13      * 0.528 3.17, 19.01 2.79e-14
##      p[HF]<.05
## 1          *
## 2          *
```

Since the Trt by Time interaction is significant, we compare the treatment effect at each time point.

```
# Effect of treatment at each time point
one_way <- df_long %>%
  group_by(time) %>%
  anova_test(dv = height, wid = unit_id, between = Trt) %>%
  get_anova_table() %>%
  adjust_pvalue(method = "bonferroni")

# Pairwise comparisons between treatments at each time point
pwc <- df_long %>%
  group_by(time) %>%
  pairwise_t_test(height ~ Trt, p.adjust.method = "bonferroni")

print(pwc)
```

```
## # A tibble: 12 x 10
##   time   .y.  group1 group2   n1   n2      p p.signif    p.adj p.adj.signif
## * <fct> <chr> <chr>  <chr>  <int> <int>   <dbl> <chr>      <dbl> <chr>
## 1 HT15  heig~ 1      2      5    5 8.06e- 5 ****    2.42e- 4 ***
## 2 HT15  heig~ 1      3      5    5 2.38e- 9 ****    7.15e- 9 ****
## 3 HT15  heig~ 2      3      5    5 4.35e- 7 ****    1.3 e- 6 ****
## 4 HT30  heig~ 1      2      5    5 7.92e- 7 ****    2.38e- 6 ****
## 5 HT30  heig~ 1      3      5    5 2.74e-11 ****    8.23e-11 ****
## 6 HT30  heig~ 2      3      5    5 1.10e- 8 ****    3.30e- 8 ****
## 7 HT45  heig~ 1      2      5    5 2.89e- 3 **      8.68e- 3 **
## 8 HT45  heig~ 1      3      5    5 1.26e-10 ****    3.79e-10 ****
## 9 HT45  heig~ 2      3      5    5 1.35e- 9 ****    4.06e- 9 ****
## 10 HT60 heig~ 1      2      5    5 1.02e- 4 ***     3.06e- 4 ***
## 11 HT60 heig~ 1      3      5    5 8.51e-11 ****    2.55e-10 ****
## 12 HT60 heig~ 2      3      5    5 3.39e- 9 ****    1.02e- 8 ****
```

We can also visualize the results as follows.

```
library(ggpubr)

ggline(df_long,
       x = "time",
       y = "height",
       color = "Trt",
       add = c("mean_se", "dotplot"),
       palette = "jco",
       ylab = "Plant Height",
       xlab = "Time (Days After Planting)",
       title = "Growth Response by Treatment")
```

